Sahaj Bhadja

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Summary

Early-career software engineer building **reliable APIs**, distributed workflows, payment systems, and data-intensive applications with **Python**, **Java**, JavaScript, SQL, cloud infrastructure, and rigorous automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Database Management Systems, Cloud Computing, Web Technologies

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous orchestration using FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ jobs while increasing throughput threefold under concurrent production workloads with reliable recovery.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Designed a concurrent batch-processing pipeline using multithreading and secure file ingestion through AWS RDS and **REST APIs**, reliably processing 1K+ files per batch under peak workloads at scale.
- Implemented an AWS-hosted backend serving 150+ instructors and 10K+ daily requests, using CI/CD-driven workflows, production monitoring, and usage tracking to reduce feedback turnaround time by 60% under load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built an end-to-end Python evaluation pipeline processing 100K+ trajectory logs, improving long-horizon planning by 18% through structured modeling, repeatable experiments, testing, and detailed simulation analysis at scale.
- Architected a spatial-temporal encoder integrating map geometry and agent histories through multi-head attention, improving closed-loop stability by 15% while validating complex behavior across simulation scenarios at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Developed a MERN resource-management platform with WebSockets, **REST APIs**, and eight microservices supporting authentication, billing, customer management, notifications, and synchronization of 2M+ records across locations in production.
- Automated Docker-based builds, testing, and staged deployments using Jenkins CI/CD, reducing deployment time by 50% while maintaining modular services and repeatable production-release workflows with automated quality gates.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Integrated multi-currency payment infrastructure supporting cards, wallets, real-time analytics, and webhook-driven refunds and disputes, reducing manual payment handling by approximately 70% across transaction workflows in production systems.
- Developed Flask APIs and Firebase workflows for 50K+ booking records, delivering sub-200-millisecond responses, real-time transaction-state synchronization, and resilient pricing and coupon validation across platforms with consistent integrity.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, HTTP/S, authentication, webhooks, microservices, multithreading, OOP
Data: PostgreSQL, SQL Server, Azure SQL, MySQL, SQLite, MongoDB, Firebase, relational schemas, query optimization, reconciliation
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux
Engineering: SDLC, idempotency, unit testing, integration testing, code review, debugging, monitoring, performance tuning, Agile/Scrum, technical documentation

Projects

PayFlowGuard | Python, SQLite, idempotency, double-entry ledger, signed webhooks
- Built and tested a payment-reliability service with idempotent retries, atomic double-entry ledger writes, signed webhook deduplication, conflicting-request validation, and currency-level reconciliation across six automated scenarios end to end.

FinNexus | Live | Next.js, TypeScript, Kubernetes, MongoDB, Plaid
- Built a scalable fintech platform supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based security, real-time synchronization, advanced filtering, and Kubernetes-backed CI/CD deployment at scale.

Chat2SQL | Code | Java, TypeScript, SQL, Docker
- Worked across a large Java and TypeScript codebase supporting 10+ databases, schema-aware query generation, structured validation, and Docker-based cross-platform deployment for unified relational data management at scale.